

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2		
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Section / Batch:	2023	Date:	

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:		100 Marks	

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.
- (a) Draw a snowflake schema diagram for the data warehouse.
- (b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?
- (c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

Roll-up: Summarize Inventory turnover by product category across all store for last quarter.

Roll up is OLAP (Online Analysis Processing) operation that is used to summarize details data into high level of information. It aggregates data by moving up towards hierarchy, making it easier to understand overall business performance instead of focusing on individual records.

In scenario the goals to summarize the inventory turnover of product based on product categories across all store for last quarter. Inventory turnover is measure of how efficiently inventory is sold and replaced during a specific period.

for example:

- Electronics - Inventory turnover: 6.8
- Groceries - Inventory turnover: 12.5
- Home Appliance - Inventory turnover: 4.3.
- Clothing - Inventory turnover: 5.2
- Furniture - Inventory Turnover: 2.9

Advantage:

- provides a high-level summary of data.
- Help management makes strategic decision.
- Reduce the complexity of detail record
- makes comparison between product categories easier

2. Drill-down: Focus on sales data for electronics in specific region, analyzed by subcategory and SKU.

> Drill down is an OLAP operation that allows users to move from summarized information to more detailed data. It helps managers investigate specific areas by breaking down data into smaller levels.

The summarized sales data

electronic categories is analysis in specific region such as south India. Instead of viewing only the total sales of electronic, data is further divided into subcategories like smartphone, laptop, TV.

example:-

phones:

- Galaxy S25 = ₹12,50,000
- iPhone 16 = ₹18,20,000
- OnePlus 13 = ₹8,60,000

laptops

- Dell = ₹9,50,000
- MacBook = ₹15,40,000
- HP = ₹7,80,000

Advantage

- provides details business insights
- help identify top-selling & low selling
- support better inventory planning

ie: Isolate sales revenue data for male customer in bronze membership across all stores

ie is OLAP that selects a single value from one dimension of cube data while keeping the remaining dimension unchanged. It allows used to focus on specific part of data.

The objective to analysis sales revenue generated by male customers who belongs to bronze membership tier across all stores

The complete sales database may include:

- Male & female customers
- bronze, silver, gold & platinum membership
- multiple product categories

Example

Store	Customer gender	membership	Sales revenue
Chennai	Male	Bronze	₹ 1,25,000
Bengaluru	Male	Bronze	₹ 1,80,000

Advantage

- simplifies data analysis
- focuses on specific business segment
- helps to understand customer behaviours

Dice: Analyse sales revenue and unit sold for selected cities & product categories during specific time period.

> Dice is OLAP operation that filter data by selecting multiple value from multiple dimension simultaneously. Unlike the slice operation which selects only one value from one dimension, dice allows selection of several value across different dimension.

In Scenario analysis focus on:

Cities: Chennai, Bengaluru.

Product Categories: Electronics & furniture

Time period: Jan to March 26

Measure: sales revenue & unit sold.

Example:

	product	Time	Sales Revenue	Unit sold
Chennai	Electronic	Jan - march	₹ 24,80,000	820
Chennai	furniture	Jan - march	₹ 8,20,000	130
Bengaluru	Electronic	Jan - march	₹ 28,10,000	910
Bengaluru	furniture	Jan - march	₹ 9,40,000	145

Advantage

- filter data using multiple dimension
- Enable details comparative analysis
- support business forecasting